

backend-swe-opt-triage.md

What This Mode Does

Triage a list of target companies for an F-1 student on OPT (active or upcoming within 12 months) targeting backend software engineering roles. Produces a ranked, tiered output using verified sponsorship history, funding signals, job liveness, and ATS detection — before any LLM is asked to interpret anything.

When to use:

- You have 10–50 companies and need to rank them by H-1B sponsorship viability before spending application time
- Your OPT window starts within 12 months and you need companies with demonstrated petition history for SWE SOC codes specifically
- You want to eliminate ghost listings and flag ATS providers before writing a single cover letter

Design rule (from engine): Run scripts and read audits first. Use an LLM only to explain or summarize after the data has been checked. Do not ask an LLM to guess sponsorship likelihood, invent labor-market statistics, or skip available audits.

Engine Layers Used

80 Days to Stay — primary. H-1B LCA disclosure data filtered to backend SWE SOC codes (15-1252 Software Developers, 15-1244 Network and Computer Systems Administrators, 15-1299 Computer Occupations All Other). Form D recency as startup viability signal.

Job-Ops — secondary. ATS provider detection, Playwright liveness checking, pipeline integrity.

Cognitive Pivot — tertiary. SOC 15-1252 cognitive demand score to flag roles resilient to AI substitution vs. facing near-term automation pressure.

Existing Data Sources and Scripts

Data Sources

Source	Path	What It Provides
80 Days company dataset	data/80days/companies.csv	30K+ companies with H-1B history and Form D signals
H-1B LCA disclosure data	data/h1b/lca_disclosure*.csv	DOL LCA records: employer, SOC code, wage, filing year
SOC occupation table	data/cognitive_pivot/soc_scores.csv	1,000+ occupations with cognitive demand scores
Form D filings	data/80days/form_d.csv	SEC Form D: company, date, amount
ATS pipeline	data/ats/pipeline.md	Current live pipeline; used to avoid duplicate entries
Application tracker	data/ats/applications.md	Existing application log; check before adding new companies

Scripts

Script	Command	What It Does
H-1B filter by SOC	python SCRIPTS/h1b_filter.py --soc 15-1252 --min-petitions 2	Filters companies by LCA petition count for given SOC
Liveness checker	node SCRIPTS/ats/liveness_check.js --url <job_url>	Returns LIVE, EXPIRED, or UNVERIFIABLE
ATS detector	python SCRIPTS/ats/ats_detector.py --company <name>	Detects Greenhouse, Lever, Ashby, or unknown
Form D recency	python SCRIPTS/form_d_filter.py --company <name> --max-age-days 548	Returns most recent Form D date and amount
Cognitive score	python SCRIPTS/cognitive_score.py --soc 15-1252	Returns cognitive demand score and AI risk tier

New Data Sources and Commands Proposed

Proposed 1 — H-1B Recency Filter

Gap: `SCRIPTS/h1b_filter.py` returns total petition count but does not weight by year. A company with 20 SWE petitions from 2018–2021 and zero since 2022 appears equal to one with 5 petitions in 2023–2024. For OPT students deciding now, recency is more predictive than lifetime count.

Proposed command:

```
python SCRIPTS/h1b_recency.py --soc 15-1252 --min-petitions 2 --since 2022
```

Status: Does not exist yet. Implementation would filter `data/h1b/lca_disclosure_*.csv` by filing year (field is present) and group by employer. No new data source required.

Proposed 2 — OPT Timeline Compatibility Check

Gap: No script checks whether a company's hiring-to-H-1B-filing lag is compatible with a specific OPT end date. Some sponsors file only after 18–24 months of employment, which may be too slow for an OPT window expiring in 12 months.

Proposed command:

```
python SCRIPTS/opt_timeline_check.py --opt-end-date YYYY-MM-DD --company <name>
```

Status: Does not exist yet, and cannot be built reliably from existing data alone.

Would require employment-start-to-LCA-filing lag per employer. Employment start date is not consistently present in LCA disclosure data. Honest gap: this script is proposed but would require supplementing with employer-reported timeline data to be reliable. Until it exists, this step requires manual research (Blind threads, recruiter email).

Step-by-Step Recipe

Follow this sequence. Do not skip to LLM interpretation before completing Steps 1–5.

Step 1 — Check existing pipeline and tracker

```
cat data/ats/pipeline.md
```

```
cat data/ats/applications.md
```

Remove from your company list any companies already in your active pipeline. Do not run triage on companies you have already applied to.

Step 2 — Run H-1B filter across SWE SOC codes

```
python SCRIPTS/h1b_filter.py --soc 15-1252,15-1244,15-1299 --min-petitions 2
```

Run across all three SWE-adjacent SOC codes. Running only 15-1252 produces false negatives for companies that file backend roles under adjacent codes.

Read the output. Record petition counts per company.

Step 3 — Run Form D recency check for pre-IPO companies only

```
python SCRIPTS/form_d_filter.py --company <name> --max-age-days 548
```

548 days = 18 months. Flag any company whose most recent Form D is older than this threshold or whose raise was under \$5M. Do not run for public companies.

Step 4 — Run liveness check for each posting URL

```
node SCRIPTS/ats/liveness_check.js --url <job_url>
```

Remove any EXPIRED or UNVERIFIABLE postings from the list before proceeding.

Step 5 — Run ATS detection

```
python SCRIPTS/ats/ats_detect.py --company <name>
```

Record ATS provider. Flag any company whose career page returns unknown or stale.

Step 6 — Run cognitive score check

```
python SCRIPTS/cognitive_score.py --soc 15-1252
```

Record score. Note: score applies to SOC code as a whole. Within-SOC role quality (distributed systems vs. CRUD) requires manual review — not verifiable from this script.

Step 7 — Assign OPT tier and record

Apply the tier definitions below. Log results to `modes/RUN_LOG.md`.

Step 8 — LLM use (optional, bounded)

After Steps 1–7 are complete and logged, you may use an LLM to:

- Draft an outreach message to a Tier 2 company asking about sponsorship intent
- Summarize what the petition count trend means for a specific company
- Draft a note explaining why a Tier 3 company was deprioritized

Do not use an LLM to guess petition counts, invent Form D data, or assign tiers

before the scripts have run.

What This Mode Can and Cannot Verify

Can Verify

- Whether a company has filed H-1B LCAs for SWE SOC codes since 2022
- Whether a company has received Form D funding in the last 18 months
- Whether a specific job posting URL is currently live
- Which ATS provider the company uses
- The cognitive demand score and AI substitution risk tier for SOC 15-1252

Cannot Verify

- **Whether the company will sponsor your specific visa situation.** LCA history is a proxy. Policy can change without appearing in the data.
- **Whether the role maps to SOC 15-1252.** Job titles are self-reported.
- **OPT timeline compatibility.** Proposed script does not exist yet.
- **Whether the ATS screens out OPT applicants.** Detection identifies provider only.
- **Startup runway.** Form D recency is a funding timestamp, not a survival signal.

Output Format

Per-Company Triage Table

Company	H-1B SWE Petitions (2022–present)	Form D Recency	ATS	Posting Live?	Cog Score	OPT Tier
Stripe	52	N/A (public)	Greenhouse	LIVE	72/100	Tier 1
Anysphere (Cursor)	0 (too new)	March 2024 – \$60M	Ashby	LIVE	72/100	Tier 2*
Wander	0	April 2023 – \$16M (FLAG)	Unknown	EXPIRED	72/100	Tier 3

OPT Tier definitions:

- **Tier 1** — Verified H-1B SWE history (≥ 2 petitions since 2022), live posting, known ATS. Apply.
 - **Tier 2** — Insufficient H-1B history (too new or too small) but strong Form D signal and live posting. Apply after manual sponsorship verification.
 - **Tier 2*** — Tier 2 with a flag: Form D age or raise amount raises runway concern. Verify sponsorship intent and viability before applying.
 - **Tier 3** — Expired posting, no recent H-1B history, no funding signal. Deprioritize unless strong referral.
-

Log Template for modes/RUN_LOG.md

```
## Run Log Entry

**Mode:** backend-swe-opt-triage.md
**Date:** YYYY-MM-DD
**OPT End Date (estimated):** YYYY-MM-DD
**SOC Target:** 15-1252 + 15-1244 + 15-1299

### Input
- Companies evaluated: [n]
- Source of company list: [LinkedIn / referral / job board]
- Checked against data/ats/pipeline.md: [yes / no]
- Checked against data/ats/applications.md: [yes / no]

### Scripts Run
- [ ] SCRIPTS/h1b_filter.py
- [ ] SCRIPTS/form_d_filter.py
- [ ] SCRIPTS/ats/liveness_check.js
- [ ] SCRIPTS/ats/ats_detect.py
- [ ] SCRIPTS/cognitive_score.py

### Output Summary
- Tier 1: [n]
- Tier 2: [n]
- Tier 2* (flagged): [n]
- Tier 3 / deprioritized: [n]
- Removed (already in pipeline): [n]
- Removed (expired posting): [n]

### Verified vs. Inferred
- VERIFIED: H-1B petition counts (LCA disclosure data)
- VERIFIED: Form D recency (SEC filings)
- VERIFIED: Job posting liveness (Playwright)
- VERIFIED: ATS provider (career page detection)
- INFERRED: OPT timeline compatibility (manual research required)
- INFERRED: Whether ATS screens OPT applicants at screening stage
```

What Changed

[How your application list changed as a result – e.g., removed 3 Tier 3 companies, added manual verification step for 2 Tier 2* companies before applying]

Flags

[Anomalies – e.g., company shows LCA history but petition count is declining year over year, or Blind threads suggest policy change not yet visible in LCA data]

Next Step

[What to check or run next]